

Assignment 4 – Pokemon Analysis

1. Distance Function [3 Points]

Study the dataset `pokedex.csv` (in Moodle) and think about a useful distance function for the Pokemon. Implement this distance function in Python and apply it to generate a pairwise distance matrix. The distance function can be based on any attributes of the Pokemon, such as their types, stats, and other relevant features (even the image thumbnails). Since all subsequent steps depend on the distance function, it is important to carefully design and implement it.

2. Clustering [3 Points]

Choose a suitable clustering algorithm (or multiple options) and cluster the Pokemon using the distance matrix generated in the previous step. Find appropriate parameters for the clustering algorithm and justify your choices (e.g. a silhouette score).

3. Visualization [3 Points]

Generate a 2d visualization using tSNE and UMAP. If you found a good solution to the clustering problem, color the points using the cluster labels you obtained in the previous step. Try to find useful parameter settings for tSNE and UMAP to get a good visualization.

4. Outlier Detection [3 Points]

Identify outliers in the dataset using an appropriate method (e.g. based on LOF, Isolation Forest, or using noise points from HDBSCAN). Are there particular characteristics that make these Pokemon outliers?

5. Discussion and Reflection [1 Points]

Discuss your findings and reflect on the process. What worked well? What challenges did you face? How would you improve your analysis if you had more time or resources?